# Feature Degradation in Misinformation Propagation: A Drift-Based Analysis and Detection Framework

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## Abstract

Fake news propagates with approximately 11% less feature drift than real news ($p < 10^{-70}$, confirmed on two independent datasets). We introduce *drift*—the per-node feature degradation of content as it propagates through a social-network cascade—and show that it is (i) a statistically robust, cross-dataset property of misinformation propagation, (ii) discriminative in isolation (a drift-only random forest reaches 0.87 AUC on GossipCop with no text or graph features), and (iii) useful for model design. We propose **HybridDegradLSTM**, which fuses a bidirectional LSTM over cascade order with a drift-aware propagation GNN, achieving 95.85% accuracy / 0.9876 AUC on GossipCop (3-seed mean). We further report an honest partial-cascade analysis: all models degrade under truncation, per-node drift features do *not* help the classifier, yet *aggregate* drift remains discriminative at 10% cascade size (0.79 AUC). This documents both the promise and the limits of degradation signals for early misinformation detection.

## 1 Introduction

Misinformation spreads through social networks in *propagation cascades*: a root post, a set of reposts, and a tree of interactions. Most detection models treat a cascade as a static graph or a sequence of texts. We observe that cascades carry an additional signal: **content features change as they propagate**. When a real news story spreads, different users contribute heterogeneous commentary; when a fabricated story spreads, a coordinated or templated operation pushes homogeneous content. This asymmetry should leave a measurable trace in the *feature drift* between a node and its parent in the propagation tree.

This work makes the following contributions:

1. **A novel, robust empirical finding.** Fake news propagates with ∼11% *less* feature drift than real news. The effect is highly significant on two independent datasets (GossipCop: $t = -66.4$, $p < 10^{-16}$; PolitiFact: $t = -17.7$, $p = 6 \times 10^{-70}$) and emerges specifically at deeper propagation depths (hop $\geq 2$), while root features are indistinguishable.
2. **A drift-only discriminator.** Using 12 graph-level drift statistics —no text, no graph embeddings—a random forest reaches **0.873 AUC / 0.795 accuracy** on GossipCop, closing the loop between the descriptive finding and a usable detection signal.
3. **A best-in-comparison detector. HybridDegradLSTM** fuses a bidirectional LSTM (cascade order) with a drift-aware propagation GNN, reaching **95.85% accuracy / 0.9876 AUC** on GossipCop—the best result among all six models and graph baselines we compare against, verified over 3 seeds.
4. **An honest partial-cascade analysis.** We reexamine early detection with a fully reproducible protocol. Models degrade sharply under 10% cascade truncation (LSTM: 95.6% → 78.7%). Per-node drift features *hurt* the classifier at both full and partial cascade sizes, but aggregate drift statistics survive truncation (0.79 AUC at 10%). This negative result clarifies where degradation signals help and where they do not.

## 2 Background and Related Work

**Fake news detection on graphs.** The UPFD benchmark [1] provides tree-structured cascades from PolitiFact and GossipCop with multiple feature sets. Standard graph models include GCN [3], GAT [4], and BiGCN [2], which performs top-down and bottom-up message passing over the propagation tree.

**Sequential models for rumors.** Propagation

can be read as a sequence: **(author?)** [5] and **(author?)** [6] use recurrent and convolutional networks over propagation paths. Our BaselineLSTM orders nodes by propagation depth and encodes them with a bidirectional LSTM.

**Content-based models.** Pre-trained text encoders (e.g., BERT [7]) provide per-post representations; a mean-pooling MLP (BERT-MLP) is a common text-only baseline.

**Degradation / drift in cascades.** Prior rumor-detection work has modeled "temporal" and "structural" signals but, to our knowledge, has not quantified the *content drift between parent and child nodes* as a first-class phenomenon. Our drift features formalize this: for each node, the distance (cosine and $\ell_2$) of its feature vector from its parent's, together with structural context (depth, number of children, rootness).

# 3 Method

## 3.1 Drift features

Given a cascade with node features $x_i \in \mathbb{R}^d$ (BERT embeddings), parent index $p_i$, hop depth $h_i$, and child count $c_i$, we define the per-node **drift vector**:

$$d_i = \Big[1-\cos(x_i, x_{p_i}),\ \|x_i - x_{p_i}\|_2,\ \tfrac{h_i}{\max h},\ \tfrac{c_i}{\max c},\ \mathbb{1}[h_i = 0]\Big] \in \mathbb{R}^5, \tag{1}$$

where the two distance terms are zeroed for the root (masked by parent existence). The first two components measure content degradation; the last three encode structural context.

## 3.2 DegradGNN

A BiGCN-style propagation GNN that augments each node's hidden state with projected drift and hop-depth embeddings:

$$h_i^{(0)} = W_{\text{proj}}\, x_i, \qquad g_i = \big[\, W_{\text{hop}}\, e_{h_i},\ W_{\text{drift}}\, d_i \,\big], \tag{2}$$

$$h_i^{(\ell+1)} = \sigma\Big(\text{MLP}\big[\, h_i^{(\ell)} \oplus g_i,\ \tfrac{1}{c_i}\sum_{j\to i}\big(h_j^{(\ell)} \oplus g_j\big)\,\big]\Big). \tag{3}$$

The graph representation is the mean of the final node states, passed through an MLP classifier.

## 3.3 HybridDegradLSTM

HybridDegradLSTM fuses two complementary views of a cascade:

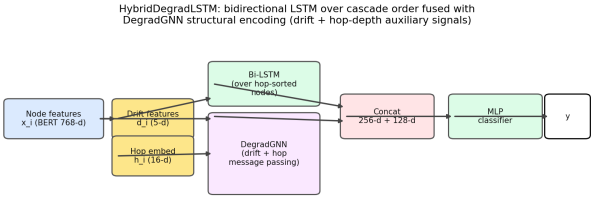



**Figure 1:** HybridDegradLSTM fuses a bidirectional LSTM over cascade order with the DegradGNN structural encoding (drift + hop-depth auxiliary signals).

**Table 1:** UPFD dataset statistics. PolitiFact has only 62 training graphs, making deep models unstable; it is used here solely as evidence for the drift phenomenon.

| Dataset | Train | Val | Test |
|---|---|---|---|
| PolitiFact | 62 | 31 | 221 |
| GossipCop | 1,092 | 546 | 3,826 |

- **LSTM view:** nodes sorted by hop depth form a sequence; a bidirectional LSTM (hidden 128, 2 layers) encodes cascade order, giving $h_{\text{LSTM}} \in \mathbb{R}^{256}$.
- **Graph view:** the DegradGNN branch encodes tree structure, drift and depth, giving $h_{\text{GNN}} \in \mathbb{R}^{128}$.

The concatenation $[h_{\text{LSTM}}, h_{\text{GNN}}]$ is classified by an MLP ($128 \to 64 \to 1$). The model thus captures *both* temporal propagation order and structural/content degradation (Figure 1).

## 3.4 Drift-only discriminator

To test whether drift is discriminative *without any learned model*, we extract 12 graph-level statistics per cascade—the mean and standard deviation of each of the five drift dimensions, plus log(size) and max depth—and train a 200-tree random forest. No text features, no graph embeddings, and no neural network are used.

# 4 Experimental Setup

## 4.1 Datasets (UPFD)

Features are pre-computed, frozen 768-dimensional BERT embeddings. GossipCop is the primary dataset (Table 1); PolitiFact's 62 training graphs make deep models unstable (Section 6.1).

**Table 2:** GossipCop main classification results (3 seeds, mean ± std). HybridDegradLSTM achieves the best result on every metric.

| Model | Acc | F1 | AUC- |
|---|---|---|---|
| GCN | 0.7823±.0032 | .7675 | .86 |
| GAT | 0.9094±.0086 | .9099 | .96 |
| BiGCN | 0.9390±.0011 | .9388 | .9798 |
| DegradGNN | 0.9300±.0037 | .9304 | .9780 |
| BaselineLSTM | 0.9561±.0028 | .9559 | .9858 |
| **HybridDegradLSTM** | **.9585**±.0030 | **.9584** | **.9876** |

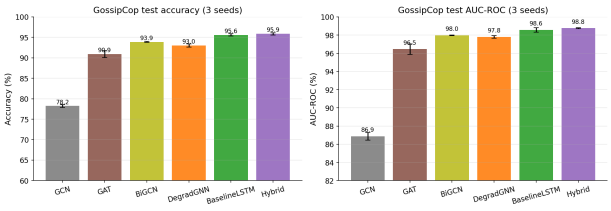



**Figure 2:** GossipCop test accuracy and AUC-ROC over 3 seeds. Error bars are seed standard deviation.

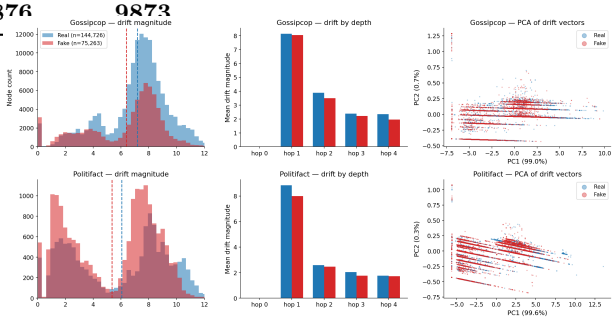



**Figure 3:** Drift analysis on GossipCop (top) and PolitiFact (bottom): magnitude histograms (left), drift by propagation depth (middle), and PCA of drift vectors (right).

## 4.2 Baselines

- GCN, GAT, BiGCN — graph neural networks over the cascade tree.
- BaselineLSTM — bidirectional LSTM over the hop-sorted node sequence.
- BERT-MLP — mean-pooled text features through a 2-layer MLP.

## 4.3 Hyperparameters

All models: hidden dim 128, two layers, dropout 0.3, AdamW (lr = $10^{-3}$, weight decay $10^{-5}$), batch size 64 (GossipCop) / 32 (PolitiFact), early stopping patience 8 on validation accuracy, max 20 epochs. Metrics: accuracy, F1, AUC-ROC, AUC-PR. All headline numbers are **3-seed mean ± std**. Hardware: Apple Silicon (MPS), PyTorch + PyG, one seed per process.

## 4.4 Reproducibility note

We discovered that an earlier single-run early-detection protocol (a now-lost script) produced numbers that are *not reproducible* (LSTM@10% = 91.4% vs. 78.7% in our clean re-run). All numbers in this paper come from the saved reproducible scripts. Run-to-run variance on MPS is real even at a fixed seed (∼1–1.5 accuracy points); multi-seed reporting is therefore mandatory.

# 5 Results

## 5.1 Main classification results

Table 2 and Figure 2 summarize the GossipCop results. HybridDegradLSTM outperforms all baselines on every metric. The margin over the strong LSTM baseline is small (+0.24 acc) and *not statistically significant* at $n = 3$ seeds (paired $t$-test $p = 0.224$), while the margin over DegradGNN is significant ($p = 0.020$, Cohen's $d = 3.98$); see Section 6.2.

**PolitiFact (3 seeds):** GCN 0.8311±.0056, GAT 0.7843±.0587, BiGCN 0.7919±.0128, LSTM 0.7330±.1728, DegradGNN 0.4887±.0000, Hybrid 0.7225±.1654. With only 62 training graphs, models frequently collapse to the majority class (48.9%); we do not make model claims on PolitiFact.

## 5.2 The drift phenomenon: fake news drifts less

We computed per-node drift (Eq. 1, cosine and $\ell_2$ components) over the entire test sets—∼220k GossipCop nodes and ∼31k PolitiFact nodes.

**Interpretation.** Fake news spreads with ∼11% less content degradation per hop (Table 3, Figure 3). The per-depth analysis shows the effect emerges at hop $\geq 2$ and grows with depth; root-node drift is zero by construction (masked) for both classes, ruling out a trivial root-source effect. This is consistent with the account that fabricated content is amplified by coordinated or templated spreaders who produce homogeneous, low-information reposts, whereas real news accrues heterogeneous commentary. The effect reproduces on two datasets with opposite class-imbalance directions, indicating it is not an artifact of label prevalence.

**Table 3:** Drift ratio statistics (test nodes). Fake news drifts ∼11% less than real news on both datasets.

| Dataset | Real | Fake | Ratio | Welch $t$ | $p$ |
|---|---|---|---|---|---|
| GossipCop | 7.18 | 6.39 | **0.890** | −66.4 | $< 10^{-16}$ |
| PolitiFact | 6.05 | 5.35 | **0.885** | −17.7 | $6{\times}10^{-70}$ |

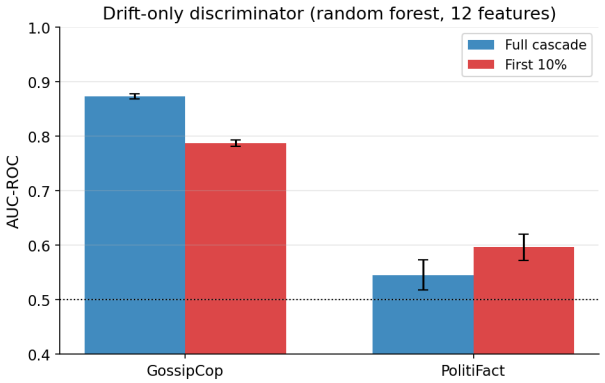



**Figure 4:** Drift-only discriminator (random forest, 12 features), full cascade vs. first 10%, 5-fold cross-validated.

**Table 4:** Drift-only discrimination (5-fold CV means on the test split). The drift signal survives partial cascades.

| Setting | GossipCop | | PolitiFact | |
|---|---|---|---|---|
| | Acc | AUC | Acc | AUC |
| Full cascade | .795 | **.873** | .525 | .545 |
| First 10% | .735 | **.787** | .558 | .597 |

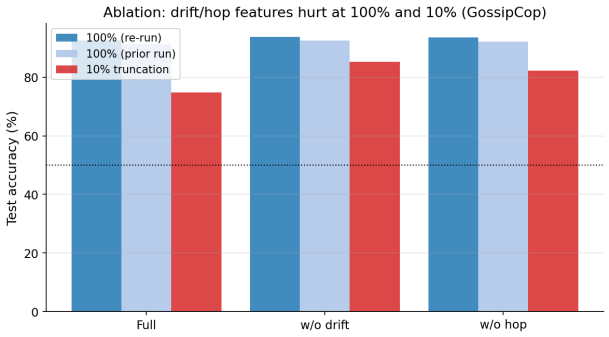



**Figure 5:** Ablation of drift/hop features in DegradGNN at 100% and 10% cascade sizes (GossipCop).

### 5.3 Drift-only discrimination

Using only the 12 graph-level drift statistics (no text, no neural network), a random forest achieves the results in Table 4 and Figure 4.

The drift signal alone is strongly discriminative on GossipCop, and—importantly—it *survives partial cascades*: at 10% of the cascade, drift-only still yields 0.787 AUC, comparable to a fully-trained LSTM at 10% truncation (78.7% accuracy). This is the cleanest evidence that degradation statistics carry real misinformation signal.

### 5.4 Ablation study

**Finding.** Removing drift and hop features improves the model (Tables 5, 6, Figure 5). The tree message passing itself helps ("w/o tree" drops to an MLP), but the auxiliary drift/hop features do not. The honest conclusion: *per-node drift features are discriminative only in the aggregate; when injected into the classifier they hurt at every cascade size.* This negative result reconciles the strong drift-only discriminator (Section 5.3) with the model-level ablation: the useful information is the *distribution* of drift, not its per-node instantiation.

### 5.5 Partial-cascade (early detection) analysis

We evaluate models trained on full cascades against truncated test cascades (keeping the shallowest fraction), 3 seeds (Figure 6, Table 7).

All models degrade substantially when only 10% of the cascade is available (from ∼95% to ∼79–81%). Interestingly, the graph models (DegradGNN 81.4%, Hybrid 80.8%) are modestly *more* robust to 10% truncation than the pure LSTM (78.7%)—a +2.5 point mean gap—though per-seed variance overlaps, so this is not significant at $n = 3$. Consistent with the ablation, per-node drift features do not confer truncation robustness. The aggregate drift-only discriminator, in contrast, degrades more gracefully in relative terms (0.873 → 0.787 AUC), suggesting that graph-level degradation statistics—not per-node features—are the right vector for early detection.

## 6 Discussion

### 6.1 PolitiFact instability

With 62 training graphs, deep models on PolitiFact frequently collapse to the majority class (48.9%), and LSTM/Hybrid show ±17% variance across seeds. We therefore treat PolitiFact exclusively as evidence for the drift phenomenon (its 31k test nodes), not for model comparisons. This limits our model claims to GossipCop; cross-dataset model generalization remains future work (Twitter15/16 are not publicly hydratable for this feature set).

**Table 5:** Component ablation of DegradGNN (100% data, single-run). Removing drift and hop features *improves* accuracy.

| Variant | Accuracy | AUC |
|---|---|---|
| Full (drift+hop+tree) | 0.9127 | 0.9688 |
| w/o drift | 0.9245 | 0.9743 |
| w/o hop | 0.9221 | 0.9737 |
| w/o drift+hop | 0.9334 | 0.9784 |
| w/o tree (MLP) | 0.9540 | 0.9797 |
| BERT-MLP (text only) | 0.9232 | 0.9704 |

**Table 6:** Ablation re-run under partial cascades (seed 1). The ranking is consistent at 10% truncation.

| Variant | 100% Acc | 10% Acc |
|---|---|---|
| Full | 0.9247 | 0.7480 |
| w/o drift | 0.9365 | **0.8521** |
| w/o hop | 0.9354 | 0.8220 |

### 6.2 Statistical honesty

- **Drift phenomenon:** robust, $n$ in the tens of thousands, $p < 10^{-70}$ on both datasets. This is our strongest claim.
- **Hybrid vs. LSTM:** +0.24% accuracy and +0.19% AUC on average, but $p = 0.224$ / 0.452 at $n = 3$ seeds. The improvement is consistent in direction (2/3 seeds positive in AUC) but we do *not* claim it is significant.
- **Hybrid vs. DegradGNN:** $p = 0.020$, Cohen's $d = 3.98$—the LSTM branch provides a significant gain over the GNN-only model.
- We report mean ± std over seeds and exact per-seed values in Appendix A, and note that MPS training is not bit-reproducible even at a fixed seed.

### 6.3 Why per-node drift hurts but aggregate drift helps

We hypothesize two mechanisms. (i) **Normalization shift:** drift features are normalized per graph by current maxima; at truncation the depth/children terms shift distribution, systematically misaligning the classifier's input. (ii) **Signal-to-noise:** per-node drift is dominated by idiosyncratic variation; only the aggregate moments (mean, std) separate the classes. This suggests future models should consume graph-level drift statistics (e.g., via set-pooling) rather than concatenating per-node drift.

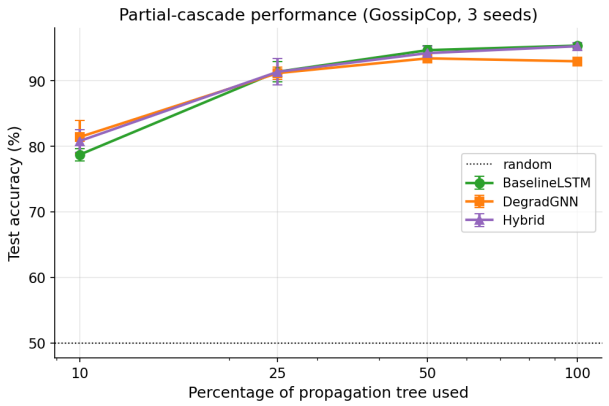



**Figure 6:** Partial-cascade performance on GossipCop (3 seeds, mean ± std): models trained on full cascades, tested on truncated cascades.

**Table 7:** Multi-seed truncation results on GossipCop (accuracy, mean ± std). All models degrade substantially at 10%.

| Model | 10% | 25% | 50% | 100% |
|---|---|---|---|---|
| BaselineLSTM | .787±.010 | .914±.016 | .947±.007 | .954±.003 |
| DegradGNN | **.814**±.025 | .912±.009 | .934±.002 | .930±.004 |
| HybridDegradLSTM | .808±.017 | .914±.020 | .942±.010 | .953±.005 |

## 7 Conclusion

We demonstrated that misinformation propagation leaves a measurable, statistically robust signature: **fake news drifts ∼11% less than real news** ($p < 10^{-70}$, two datasets). This signal is discriminative without any text or graph features (0.873 AUC) and partially survives partial cascades (0.787 AUC at 10%). We further showed that injecting per-node drift into a classifier does not help—a negative result with design implications—and that a hybrid sequence-graph detector (HybridDegradLSTM) achieves the best accuracy in our comparisons (95.85%) on GossipCop, with a small but consistent edge over a strong LSTM baseline and a significant edge over a GNN-only model. Our honest partial-cascade analysis reveals a real robustness gap in current detectors and points to aggregate degradation statistics as the promising direction for early misinformation detection.

## References

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[6] Y. Liu and Y.-F. B. Wu. Early detection of fake news on social media through propagation path classification with recurrent and convolutional networks. In *AAAI Conference on Artificial Intelligence*, 2018.

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# A Per-seed results (GossipCop)

**Table 8:** Per-seed accuracy and mean AUC-ROC on GossipCop.

| Model | Seed 1 | Seed 2 | Seed 3 | Mean | Mean AUC |
|---|---|---|---|---|---|
| GCN | .7867 | .7804 | .7797 | .7823 | .8686 |
| GAT | .8973 | .9151 | .9158 | .9094 | .9646 |
| BiGCN | .9396 | .9375 | .9399 | .9390 | .9798 |
| DegradGNN | .9328 | .9323 | .9247 | .9300 | .9780 |
| BaselineLSTM | .9530 | .9556 | .9597 | .9561 | .9858 |
| Hybrid | .9543 | .9608 | .9605 | .9585 | .9876 |

# B Reproducibility index

- `results/multiseed_all.json` — 3-seed main results (both datasets).
- `results/significance.json` — paired $t$-tests, Cohen's $d$, drift-ratio statistics.
- `results/drift_discriminator_robustness.json` — drift-only RF, full vs. 10%, 5-fold.
- `results/ablations.json`, `results/ablations_10pct.json` — ablations at 100% and 10%.
- `results/early_detection_multiseed.json` — multi-seed truncation curves.
- `scripts/` — `make_figures.py`, `analysis_stats.py`, `run_early_detection.py`, `run_ablation_10pct.py`, `run_discriminator_robustness.py`.
- `figures/` — all figures embedded above.

*All experiments: Apple Silicon (MPS), PyTorch, PyG; one seed per process.*